AI Health &
Wellness Coach

User Guide & Manual

Version 1.0 | MERN Stack + OpenAI

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1. Introduction

1.1 About This Application

The AI Health & Wellness Coach is a full-stack web application built on the MERN stack (MongoDB, Express.js, React, Node.js) with OpenAI API integration. It provides users with a personalised, AI-driven health companion that generates custom meal plans, workout routines, and real-time coaching — all tailored to each user's individual profile, goals, and progress history.

1.2 Who This Guide Is For

This guide is intended for end users of the application. No technical knowledge is required. Each section walks through a specific feature with step-by-step instructions, tips, and explanations.

1.3 Key Features at a Glance

Feature	What It Does
AI Meal Planner	Generates a full daily meal plan based on your calorie goal and dietary needs
Workout Generator	Creates personalised workout plans with progressive overload
AI Coach Chat	A real-time streaming chatbot that knows your stats and history
Progress Tracker	Logs weight, mood, sleep, and workouts with visual charts
Symptom Checker	Provides general wellness information with safety disclaimers
Daily Insights	AI-generated personalised summary of your weekly trends

2. Getting Started

2.1 Creating Your Account

To begin using the application, you must first create a personal account. Your account stores all your health data, preferences, and AI conversation history securely.

Step-by-step registration

1. Open the application in your browser and click Get Started or Sign Up.
2. Enter your full name, a valid email address, and a password (minimum 8 characters).
3. Click Create Account. You will be directed to the Onboarding form.

2.2 Completing Your Health Profile (Onboarding)

The onboarding form is the most important step. The information you provide here is used as context for every AI response you receive. The more accurate your details, the better the AI can personalise recommendations.

Information required

- Age and biological sex
- Current weight (kg) and height (cm)
- Primary fitness goal: Lose Weight, Gain Muscle, or Maintain Weight
- Activity level: Sedentary, Lightly Active, Moderately Active, or Very Active
- Dietary restrictions (e.g. vegetarian, vegan, gluten-free, dairy-free)

Tip
You can update your profile at any time from the Profile Settings page. The AI will automatically use your updated details on the next request.

2.3 Dashboard Overview

After completing onboarding, you land on your personal Dashboard. This is your home screen and shows:

- Today's calorie intake vs. your daily target
- Macronutrient breakdown (protein, carbs, fat)
- Water intake log

- Workout completion status for today
- A personalised AI daily greeting and summary
- Quick-access buttons to all major features

3. AI Meal Planner

The Meal Planner uses OpenAI to generate a complete daily meal plan tailored specifically to your calorie target, macronutrient goals, and dietary restrictions.

3.1 Generating a Meal Plan

4. Navigate to the Meal Planner page from the sidebar or dashboard.
5. Click the Generate Meal Plan button.
6. The AI will stream your meal plan in real-time. You will see meal cards appear one by one as the AI generates them.
7. Each card shows the meal name, calories, protein, carbs, and fat.
8. A macro doughnut chart updates automatically once the full plan is loaded.

3.2 Understanding Your Meal Plan

Meal Type	Typical Timing	Purpose
Breakfast	7:00 AM - 9:00 AM	Kickstart metabolism, energy for the morning
Lunch	12:00 PM - 1:30 PM	Main fuel meal, complex carbs and protein
Dinner	6:00 PM - 8:00 PM	Recovery meal, lean protein and vegetables
Snacks	Between meals	Keep energy stable, prevent overeating at meals

3.3 Logging Meals

To log a meal you have eaten, click the Log This Meal button on any meal card. The meal and its macros are saved to your daily log and immediately reflected on the Dashboard calorie counter.

Important

Your daily calorie target is calculated automatically based on your weight, height, age, activity level, and goal. The AI is instructed to stay within 50 kcal of this target when generating your plan. If you want to adjust your target, update your profile.

3.4 Regenerating or Customising

- Click Generate Meal Plan again at any time to create a fresh plan.
- To request specific changes (e.g. 'no eggs', 'higher protein'), use the AI Coach Chat and describe what you need.

4. AI Workout Planner

The Workout Planner generates personalised workout routines based on your fitness goal, available equipment, and previous workout history. The AI applies progressive overload automatically — meaning it will suggest slightly heavier weights or more reps each session to keep you progressing.

4.1 Generating a Workout

9. Go to the Workout Planner page.
10. Select your preferences using the form: workout type, duration, and available equipment.
11. Click Generate Workout.
12. Your workout plan will appear as a list of exercises with sets, reps, rest time, and coaching notes.

4.2 Workout Types Available

- Strength Training — barbell and dumbbell exercises for muscle building
- Cardio — running, cycling, jump rope, and HIIT intervals
- Yoga & Flexibility — guided pose sequences with hold durations
- HIIT — high-intensity interval training for fat loss and conditioning
- Bodyweight — no-equipment workouts suitable for home use

4.3 Reordering Exercises

You can drag and drop exercises within the workout list to reorder them before starting. This is useful if you want to prioritise certain muscle groups or prefer a different warm-up sequence.

4.4 Logging a Completed Workout

13. For each exercise, enter the actual sets, reps, and weight you completed.
14. Add optional notes (e.g. 'felt easy' or 'lower back tight').
15. Click Save Workout. Your session is recorded and used to improve the next AI-generated plan.

How Progressive Overload Works

When you log a workout, the app stores your exercise data. The next time the AI generates a workout, it receives your last 3 sessions as context and will suggest 2.5-5 kg increases or additional reps where appropriate. This mirrors what a personal trainer would do.

5. AI Coach Chat

The AI Coach is a real-time streaming chatbot that acts as your personal health and fitness coach. Unlike a generic AI assistant, the Coach knows your profile, your goals, and what you have eaten and done today — making every conversation highly contextual.

5.1 Starting a Conversation

16. Click AI Coach in the sidebar, or tap the Chat with Coach button on the Dashboard.
17. Type your message in the input field at the bottom and press Enter or click Send.
18. The AI Coach will respond in real-time — you will see words appear as they are generated.

5.2 What You Can Ask

- Nutrition questions: 'How much protein should I eat on rest days?'
- Workout advice: 'My lower back is sore — what should I avoid today?'
- Meal adjustments: 'I want a higher-protein meal plan for tomorrow'
- Motivation: 'I missed 3 workouts this week — how do I get back on track?'
- Progress interpretation: 'My weight went up 0.5 kg — is that bad?'
- Recipe ideas: 'What are some high-protein breakfast ideas under 400 calories?'

5.3 Context Awareness

The AI Coach automatically receives the following context with every message you send:

- Your name, age, weight, height, and fitness goal
- Your daily calorie target and how many calories you have consumed today
- Whether you have completed today's workout
- The last 10 messages in the current conversation

Tip

You do not need to re-explain who you are or what your goals are. The AI already knows. Just ask your question naturally, as you would to a coach who knows you well.

5.4 Conversation History

Your conversation history is preserved within a session. If you close the browser and return, a new session begins. To continue a previous conversation, scroll up to see past messages — they are stored in your account.

6. Progress Tracker

The Progress Tracker gives you a clear visual view of your health trends over time. It combines weight tracking, habit streaks, workout history, and AI-generated insights into a single page.

6.1 Logging a Daily Check-In

19. Go to the Progress Tracker page.
20. Click Log Today in the top right corner.
21. Enter your current weight, mood (1 to 5 scale), sleep hours last night, and optional notes.
22. Click Save Entry.

6.2 Weight Chart

Your logged weights are displayed as a line chart covering the last 30 days. The chart updates immediately each time you log an entry. The trend line shows whether your weight is moving in the direction of your goal.

6.3 Workout Habit Heatmap

The habit heatmap shows a calendar view of the past 3 months. Each day is colour-coded:

Colour	Meaning
Dark green	Workout completed and calorie goal met
Light green	Workout completed or calorie goal met (not both)
Amber	Partial activity — logged but below target
Light grey	No entry logged for that day

6.4 Current Streak

Your current streak shows the number of consecutive days you have logged an entry. The best streak is also displayed so you have a personal record to beat. Consistency is more important than perfection — even logging a rest day counts toward your streak.

6.5 AI Weekly Insight

Once per day, the application generates a 2-sentence personalised insight based on your last 30 days of progress. This insight is displayed at the top of the Progress Tracker page. It highlights a specific pattern in your data and suggests one actionable improvement.

Example Insight

You have been very consistent Monday to Thursday, but your weekend logging drops off — this may be affecting your weekly calorie average. Try setting a Saturday morning reminder to log your check-in before the day gets busy.

7. Symptom Checker

Medical Disclaimer
The Symptom Checker is NOT a medical diagnostic tool. It provides general wellness information only. Always consult a qualified healthcare professional for medical advice, diagnosis, or treatment. In an emergency, contact your local emergency services immediately.

The Symptom Checker allows you to describe how you are feeling and receive general wellness information, possible lifestyle-related causes, and practical self-care tips.

7.1 Using the Symptom Checker

23. Navigate to Symptom Checker in the sidebar.
24. Type a description of your symptoms in the text box (e.g. 'I have had a headache all day and feel tired').
25. Optionally add your current medications if relevant.
26. Click Analyse.
27. The AI will return: possible general causes, wellness tips, and a mandatory recommendation to consult a doctor.

7.2 What the AI Will and Will Not Do

The AI Will	The AI Will Not
Provide general wellness information	Diagnose any condition
Suggest hydration, rest, and lifestyle tips	Prescribe or recommend medication
Explain common causes of general symptoms	Replace professional medical advice
Always recommend seeing a doctor	Provide emergency medical guidance

8. Account Settings & Profile

8.1 Updating Your Health Profile

Your health profile can be updated at any time. Navigate to Profile in the top-right menu or sidebar. You can update:

- Current weight (recommended: update weekly for accurate AI recommendations)
- Fitness goal (if your goal changes, update this immediately)
- Activity level
- Dietary restrictions
- Profile photo

8.2 Changing Your Password

28. Go to Settings > Security.
29. Enter your current password, then your new password twice.
30. Click Update Password.

8.3 Notification Preferences

You can enable or disable the following notifications from Settings > Notifications:

- Daily check-in reminder (set your preferred time)
- Workout reminder
- Weekly progress summary email
- AI insight notification

8.4 Deleting Your Account

To permanently delete your account and all associated data, go to Settings > Account > Delete Account. This action is irreversible. All your logs, conversations, and profile data will be permanently removed.

9. Troubleshooting

Problem	Solution
AI response takes too long	Check your internet connection. If the issue persists, refresh the page and try again.
Meal plan looks wrong for my goal	Verify your calorie target in Profile Settings. Ensure your weight, height, and activity level are current.
I cannot log in	Use the Forgot Password link on the login page to reset your password via email.
My workout streak reset unexpectedly	Streaks reset if no entry is logged on a calendar day. Check that you logged a check-in yesterday.
Chat is not streaming / appears frozen	Refresh the page. Streaming requires a stable connection. Avoid using the app on very slow networks.
Progress chart is empty	The chart requires at least 2 logged entries to render a line. Log your weight for a second day.

10. Glossary

Term	Definition
MERN Stack	A set of JavaScript technologies: MongoDB, Express.js, React, and Node.js used to build full-stack web apps.
OpenAI API	The artificial intelligence service that powers all AI features in this application.
Streaming	A method of delivering AI responses token-by-token in real time, so you see words appear as they are generated.
Progressive Overload	A training principle where you gradually increase exercise difficulty over time to continue making progress.
Macros / Macronutrients	The three main food groups: protein, carbohydrates, and fat. Measured in grams per day.
Calorie Target	Your personalised daily calorie goal, automatically calculated from your profile data.
BMI	Body Mass Index — a measure of body weight relative to height. Used as a general health indicator.
System Prompt	Hidden instructions sent to the AI before your message, telling it who you are and how to behave.
JWT	JSON Web Token — a secure credential stored in your browser that keeps you logged in.

11. Support & Feedback

If you encounter an issue not covered in the Troubleshooting section, or if you have suggestions for improving the application, please use one of the following channels:

- In-app feedback: Settings > Send Feedback
- Email support: support@wellnessai.app
- GitHub issues: github.com/your-org/wellness-ai/issues

When reporting a bug, please include a brief description of the problem, the steps that led to it, and your browser and operating system. This helps the development team reproduce and fix the issue quickly.

12. Backend Development Roadmap

You have completed Register and Login. Below is the exact order to build the remaining backend — each step builds on the previous one. Follow this sequence and nothing will block you.

Phase 1 — Already Done (Register & Login)

Status: Complete

POST /api/auth/register and POST /api/auth/login are working. JWT is being saved on the frontend. This is your foundation — every protected route will use the authMiddleware you built here.

Phase 2 — User Profile

This is the most important step after auth. Every AI feature reads the user profile before building a prompt. Build this before touching any AI routes.

Step	What to Build	Practical Hint
2.1	authMiddleware.js — protect routes	export a middleware that reads Authorization: Bearer <token>, verifies the JWT, and attaches req.userId. Add this to every protected route.
2.2	User Model — add health fields	Add age, weight, height, goalType (enum: lose/gain/maintain), activityLevel, dietaryRestrictions (array), dailyCalorieTarget, profileComplete (Boolean) to your User schema.
2.3	GET /api/user/profile	Return the logged-in user's full profile. Call getProfile(req.userId) — this function will be reused by every AI service.
2.4	PUT /api/user/profile	Update weight, goal, activity, restrictions. After saving, recalculate and update dailyCalorieTarget server-side using the Harris-Benedict formula.
2.5	Calculate calorie target	Formula: BMR = 10*weight + 6.25*height - 5*age (+5 male / -161 female). Multiply by activity multiplier (1.2 to 1.9). Adjust for goal: -500 kcal to lose, +300 to gain.

Key Hint

Create a helper file services/profileService.js with a single function getProfile(userId) that fetches and returns the full user object. Call this at the top of every AI route. This one function makes all AI responses personalised.

Phase 3 — MongoDB Models (build all 3 before routes)

Build these 3 schemas now so your routes have something to write to. Do not skip ahead to routes before the models exist.

Step	What to Build	Practical Hint
3.1	MealLog Model	Fields: userId, mealType (breakfast/lunch/dinner/snack), name, calories, protein, carbs, fat, loggedAt. Add index on {userId, loggedAt} for fast daily queries.
3.2	Workout Model	Fields: userId, planName, type, durationMinutes, exercises [{name, sets, reps, weightKg, notes}], completedAt, aiGenerated (Boolean). Nested array for exercises.
3.3	ProgressLog Model	Fields: userId, date (Date only, no time), weightKg, mood (1-5), sleepHours, notes. Add unique compound index on {userId, date} to prevent duplicate daily entries.

Key Hint

For ProgressLog, use a unique index on {userId, date} and always use findOneAndUpdate with upsert: true in your route. This means logging twice in one day just updates the entry instead of throwing a duplicate error.

Phase 4 — Meal & Workout CRUD Routes

Now wire up the routes that log and fetch meals and workouts. These give you real data to feed into the AI later.

Step	What to Build	Practical Hint
4.1	POST /api/meals/log	Save a logged meal. Auto-calculate total calories server-side: (protein * 4) + (carbs * 4) + (fat * 9). Never trust the frontend to do this math.
4.2	GET /api/meals/today	Return all meals for today using date range query: {loggedAt: {$gte: startOfDay, $lte: endOfDay}}. Also sum and return total calories and macros.

4.3	POST /api/workouts/log	Save a completed workout with exercises, sets, reps, and weights. Set aiGenerated: true if the plan came from AI.
4.4	GET /api/workouts/history	Return last 10 workouts sorted by completedAt desc. This array gets injected into the AI workout prompt for progressive overload.
4.5	DELETE /api/meals/:id	Allow users to remove a meal log entry. Check that the meal belongs to req.userId before deleting — never skip ownership check.

Phase 5 — Progress Routes

These routes power the Progress Tracker page and the AI insight generator.

Step	What to Build	Practical Hint
5.1	POST /api/progress	Save daily check-in using findOneAndUpdate + upsert: true so duplicate days just update. Accept: weightKg, mood, sleepHours, notes.
5.2	GET /api/progress/chart	Return last 30 days of logs sorted by date asc, grouped by date. This feeds directly into the Recharts LineChart on the frontend.
5.3	GET /api/progress/streak	Calculate current streak: query logs sorted by date desc, count consecutive days from today backwards. Stop when a day has no entry. Return {current, best}.
5.4	GET /api/progress/summary	Return weekly averages: avg weight, workout completion %, avg mood. This compact summary gets sent to the AI insight generator.

Key Hint

For the streak, do not count all logs in one query and then loop in JS. Query with sort: {date: -1}, limit: 60, then walk backwards from today. Stop the first time you find a missing day. This is much faster than processing the entire log history.

Phase 6 — AI Routes (build last)

Do not start AI routes until Phases 2 to 5 are complete. Each AI route needs real profile data and real history to work properly. Building AI first without data gives generic, useless responses.

Step	What to Build	Practical Hint

6.1	POST /api/ai/meal-plan	Fetch user profile, build system prompt with calorie target and restrictions, call OpenAI with stream: true, pipe SSE to frontend. Return structured JSON: {breakfast, lunch, dinner, snacks}.
6.2	POST /api/ai/workout	Fetch profile + last 3 workouts, build prompt with progressive overload context, use response_format: {type: json_object} to force clean JSON. Validate with Zod before saving.
6.3	GET /api/ai/chat (SSE)	Accept userMessage + conversationHistory array. Inject profile + today stats into system prompt. Stream response as SSE: data: {token} chunks. End with data: [DONE].
6.4	GET /api/ai/insight	Fetch 30-day progress summary, send compact text to GPT (not full JSON), ask for exactly 2 sentences. Cache result in MongoDB for 24 hours — no need to regenerate every load.
6.5	POST /api/ai/symptom	Run openai.moderations.create() on input first. If flagged, return 400. Always prepend disclaimer to system prompt. Never diagnose — only return general wellness info.

Most Important AI Hint

Your system prompt is everything. Template it like this: You are a health coach for {name}. Goal: {goal}. Today: {calories}/{target} kcal eaten, workout {done/not done}. Restrictions: {list}. This single prompt makes every response feel like a real personal coach, not a generic chatbot.

Phase 7 — Final Polish Before Frontend Connect

Step	What to Build	Practical Hint
7.1	Error handling middleware	Add a global error handler in app.js: app.use((err, req, res, next) => res.status(err.status || 500).json({message: err.message})). All route errors flow here.
7.2	Input validation	Add express-validator to all POST routes. Validate email format, password length, required fields. Return 400 with clear error messages — the frontend displays these directly.
7.3	Rate limiting	npm install express-rate-limit. Apply to /api/auth/* routes: max 10 requests per 15 minutes. Apply to /api/ai/* routes: max 20 per hour to control OpenAI costs.

7.4	CORS setup	npm install cors. Allow only your Vite frontend origin: cors({origin: http://localhost:5173, credentials: true}). Without this, every frontend fetch will be blocked.
7.5	Environment variables	Move all secrets to .env: MONGO_URI, JWT_SECRET, OPENAI_API_KEY, PORT. Add .env to .gitignore immediately. Never commit API keys.

Build Order Summary

Follow this exact sequence. Each row depends on everything above it.

#	Phase	Note
1	**Register + Login**	Complete
2	**User Profile (Model + Routes)**	Build next — AI depends on this
3	**MongoDB Models (Meal, Workout, Progress)**	Build all 3 before any routes
4	**Meal + Workout CRUD routes**	Gives AI real data to use
5	**Progress routes + streak**	Powers tracker and AI insights
6	**AI routes**	Build last — needs all data ready
7	**Error handling, validation, CORS, rate limit**	Polish before connecting frontend

Golden Rule

Never build an AI route before you have real data to feed it. A blank database gives generic AI responses. A populated database with user profile, meal history, and workout logs gives responses that feel like a real personal coach knows you.

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